

**Does Teleporting Get You Lost? Comparing Joystick, Teleport, and Trail-Assisted Teleport
Movement in a VR Escape Task**

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CS464 Principles of Human-Computer Interaction

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October 5, 2026

Problem Statement

In virtual reality (VR) worlds, the bounds are larger than the user's physical space, so navigation tools are needed to explore the virtual world or complete a task. The main navigation methods are joysticks and teleportation, but each of these has a downside. For joysticks, a handheld device that lets users move, turn, and navigate, it feels smooth during actions but can cause motion sickness for users. On the other hand, teleportation, instantly moving to a new location instead of walking smoothly, doesn't cause motion sickness, but many users lose track of their position. The consequences of each method are that users either feel sick and take off the headset or waste time going back over the previous positions. This hurts the various uses of VR such as training, classes, games, and virtual tours. Right now designers have to choose between comfort and orientation. It is important to investigate whether a simple change can give users both, because every VR application that involves movement faces this trade-off.

Gap In Knowledge

Based on the peer-reviewed articles we uncovered that instantly teleporting users made them disoriented, having them search for the target again after the jump (Bowman, Koller & Hodges, 1997). Comparing joystick, teleport, and redirected walking, joystick caused the most motion sickness, and users preferred teleportation and redirected walking over the joystick (Langbehn, Lubos, & Steinicke, 2018). In a study of 26 participants, the joystick had the highest self-reported sickness, and while teleportation felt the most effective because of its speed, it broke the immersion in the virtual world (Boletsis & Cedergren, 2019). Another author has five experiments resulting in teleporting disorienting users more than walking; walls helped but did not completely fix it (Cherep et al 2020). It has also been concluded that cues in a virtual world shape how people find their way, leading us to hypothesize that a trail could help (Darken & Sibert 1996). We already know that teleportation disorients users and a joystick makes them sick. To our knowledge, no study has tested whether a simple trail fixes the teleportation problem in a search task where getting lost costs times, while also checking that it does not bring motion sickness back.

Research Questions

Objective: To find which movement method gets users through a VR escape task fastest with the least motion sickness, and whether a trail fixes teleport's habit of getting users lost.

RQ1: Does the movement method affect how efficiently people complete the escape task?

H1a: Teleport with a trail will be the fastest of the three.

H1b: Teleport with a trail will have less backtracking than the plain teleport.

RQ2: Does the movement method affect motion sickness?

H2a: Joystick will cause more motion sickness than both teleport methods.

H2b: Adding the trail won't make people sicker than plain teleport.

Proposed Experiment

Apparatus:

We will be using the Meta Quest 3 headset with an app we will build in Unity. using the XR Interaction Toolkit. It is optimal since it is standalone, so no PC is needed, and the toolkit already has joystick and teleport built in. The app logs position and time automatically.

Task & Variables

The participant starts in the first of three connected rooms, with each room hiding a lever. Flipping a lever opens the gate to the next room, and the third lever unlocks the exit. The goal is to flip all three and get out as fast as you can, with a 5-minute cap. Why? You have to search each room, so losing track of where you've looked costs time. The gates keep everyone on the same route, so nobody gets stuck like they could in a maze.

The independent variable is the movement method (3 levels): joystick, teleport, and teleport with a trail, because joystick and teleport are the two standard methods, and the trail version shows what the trail alone adds.

The dependent variables are completion time (seconds), logged by the app; backtracking, measured by the number of times someone goes back into a part of the floor they already visited based on the position log; and motion sickness, measured using the Simulator Sickness Questionnaire (SSQ; Kennedy et al., 1993) after each method. At the end, participants rank the three methods and indicate which of the three felt most comfortable.

Design and Counterbalancing

The study uses a within-subjects design, meaning that every participant tries all three methods. This reduces the number of participants needed and ensures differences between participants don't skew it. The same set of rooms is used for each trial, but three different lever layouts (A, B, C) are used so that the levers are in different spots each round and nobody can memorize them. A 3×3 Latin square rotates the order, so each method comes first, second, and third equally often and gets paired with each lever layout. We plan to recruit 9–12 participants (a multiple of 3). Each completes a baseline survey first and a 5-minute break between rounds so sickness can fade.

Real Life Application

VR training is used for safety, medical, education, games, virtual tours, and real estate walkthroughs, and all of these need people to move around without getting sick or lost. Designers are the ones who use research; they get evidence on which movement method to make the default, and whether a cheap trail feature is worth adding. First-time VR users and people who get motion sick easily benefit the most.

Relation to HCI

Movement is how the user controls a VR system, so it's a core HCI design problem. The usability trade-off is efficiency (time, backtracking) vs. comfort (motion sickness). That maps to efficiency and satisfaction in ISO 9241-11. The wayfinding aid is the trail adds information to the interface so people don't get disoriented (Darken & Sibert, 1996). The evaluation is a controlled within-subjects experiment with logged behavior plus a standard questionnaire.

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